Middle East Technical University

EE568 – Project 1 Report

Torque in a Variable Reluctance Machine

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GitHub Repository:
https://github.com/polatztrk/EE568_Project_1

Abstract

The purpose of this report is to analyse the characteristics of torque generation in a variable reluctance machine through the use of analytical models and FEA. Initially, the variation of inductances and electromagnetic torque generation will be analysed theoretically assuming a number of assumptions. Subsequently, 2D FEA will be done with a number of magnetic material models such as linear and non-linear materials. In addition, an approach for continuous rotation control is also presented.

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1 Introduction

This report presents the modelling and analysis of a variable reluctance machine. The main objectives are:

- to derive an analytical model for reluctance, inductance, and torque,
- to validate the analytical model using 2D FEA,
- to compare linear and nonlinear magnetic material models,
- to investigate the influence of fringing and saturation,
- to propose a control strategy that enables continuous positive torque production.

2 Problem Definition and Given Parameters

2.1 Machine Description

Machines consists of one C shaped stator core and one rotor with varying height as a result varying reluctance. And stator excited with the 300 turn copper coil with 2.5A.

2.2 Given Data

Table 1: Given machine parameters

Parameter	Value
Coil window dimensions	30mmx10mm
Air-gap clearance (each)	0.5mm
Core depth	25 mm
Number of turns	300
Coil current	2.5 A DC

2.3 Geometry Figures

The figures which describe dimensions and shape given in the project details can be seen below.

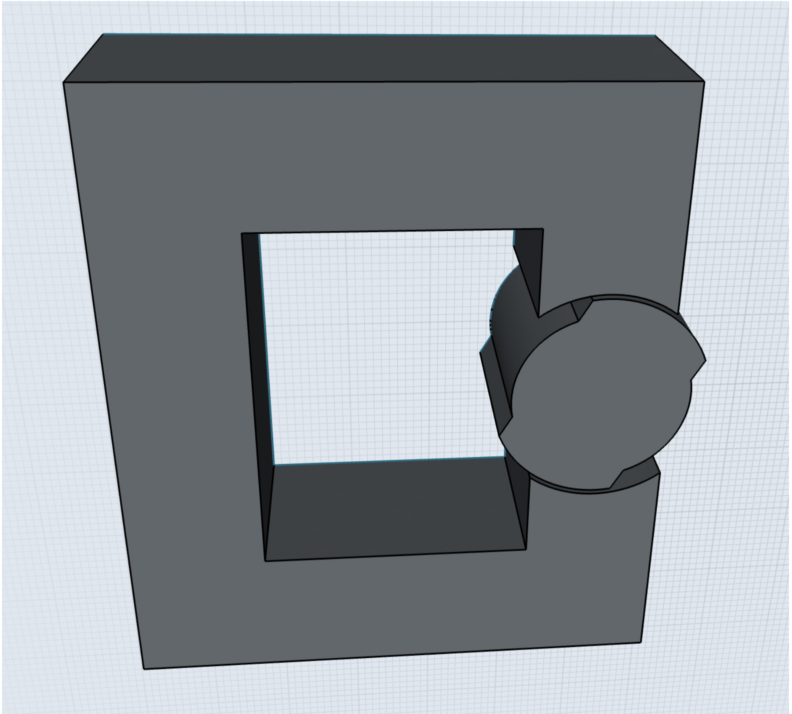

Figure 1: Variable reluctance machine geometry

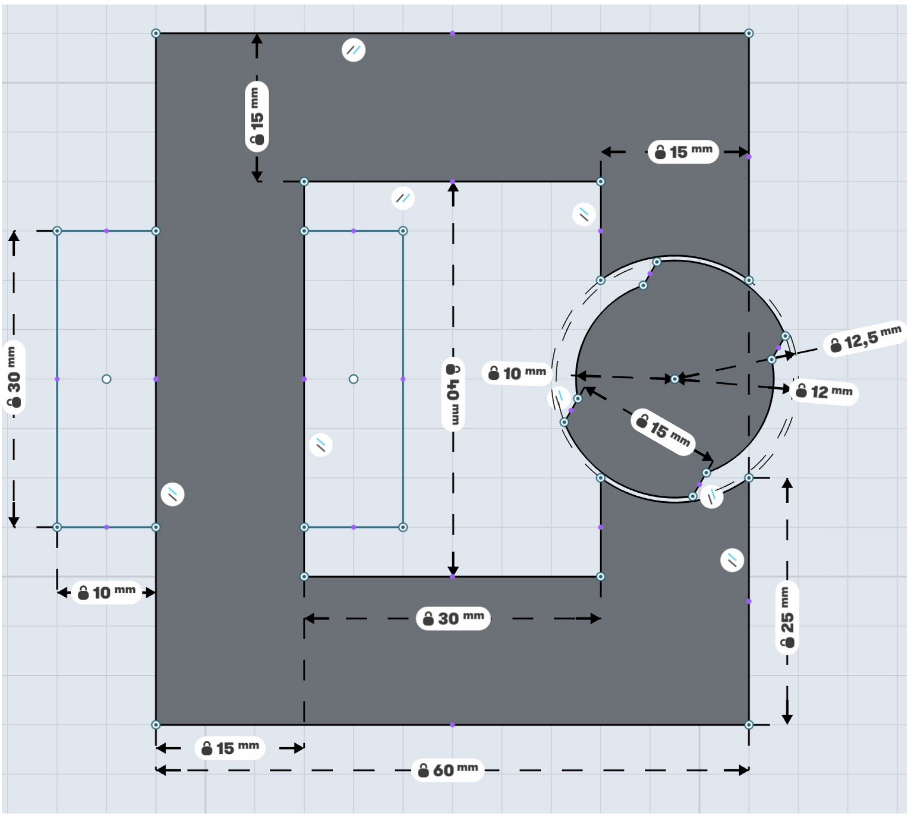

Figure 2: Machine dimensions used in the project

3 Q1 - Analytical Modelling

3.1 Part a

3.1.1 Assumptions

For this part of the project i made following assumptions:

- Core permeability is assumed infinite for the base analytical model.
- Flux leakage is neglected.
- Fringing is neglected in the first-order model.
- Inductance variation with rotor angle is approximated as sinusoidal.
- Air gap cross-section assumed rectangular.
- In rotors 90 degree position air gap cross section calculated as average.

3.1.2 Magnetic Reluctance Derivation

Since the assumption is sinusoidal approximation. I calculate the reluctance for the where it is maximum, which corresponds to 0 degree, and where it is minimum where rotor position is 90 degree. For rotor position 0 degree, I calculated air gap cross-section as:

$$A_0 = 25mm * 15mm = 375mm^2 = 3.75 * 10^{-4}m^2 \tag{1}$$

Since core permeability is assumed infinite we can directly calculate minimum reluctance using air gap. The total distance of air gap is 2 time 0.0005 mm so we can calculate like this:

$$\mathcal{R}_{min} = \frac{2 * l_{g0}}{\mu_0 A_0} = \frac{0.001}{4 * \pi * 10^{-7} * 3.75 * 10^{-4}} = 2.122 * 10^6 A/Wb \tag{2}$$

For rotor position 90 degree, rectangular cross-section of the rotor side is 24 mm*25 mm. And in core side it is 15 mm * 25 mm. So we can calculate the volume in here as using average cross section are. Which becomes:

$$A_{90} = \frac{((24mm * 25mm) + (15mm * 25mm))}{2} = \frac{600mm^2 + 375mm^2}{2} = 4.875 * 10^{-4}m^2 \tag{3}$$

The total distance of air gap is 2 times the 0.0025 mm. So maximum reluctance can be derived as:

$$\mathcal{R}_{max} = \frac{2 * l_{g90}}{\mu_0 A_9 0} = \frac{0.005}{4 * \pi * 10^{-7} * 4.875 * 10^{-4}} = 8.161 * 10^6 A/Wb \tag{4}$$

3.1.3 Inductance Derivation

Using magnetic circuit relations, we can find that maximum inductance is:

$$L_{max} = \frac{N^2}{\mathcal{R}_{min}} = \frac{300^2}{2.122 * 10^6} = 0.0424H \tag{5}$$

And the minimum inductance is:

$$L_{min} = \frac{N^2}{\mathcal{R}_{max}} = \frac{300^2}{8.161 * 10^6} = 0.011H \tag{6}$$

Now we can write angle dependent sinusoidal form inductance as:

$$L(\theta) = L_{\text{avg}} + L_{\text{amp}} \cos(2\theta) \tag{7}$$

Where,

$$L_{\text{avg}} = \frac{L_{max} + L_{min}}{2} = 0.0267H \tag{8}$$

$$L_{\text{amp}} = \frac{L_{max} - L_{min}}{2} = 0.0157H \tag{9}$$

3.1.4 Torque Derivation Under Constant DC Excitation

The magnetic energy of the system can be written as:

$$W_m(\theta) = \frac{1}{2} L(\theta) i^2 \tag{10}$$

Since the derivative of magnetic energy with respect to position. We can write this.

$$T_e(\theta) = \frac{dW_m(\theta)}{d\theta} = \frac{d(\frac{1}{2}(L_{avg} + L_{amp}\cos(2\theta))i^2)}{d\theta} = -i^2 L_{amp} \sin 2\theta \tag{11}$$

3.2 Part b - Analytical Results

We can see the 0 to 180 degree spanned results for analytical calculations of inductance and torque below.

3.2.1 Inductance Versus Angle Plot

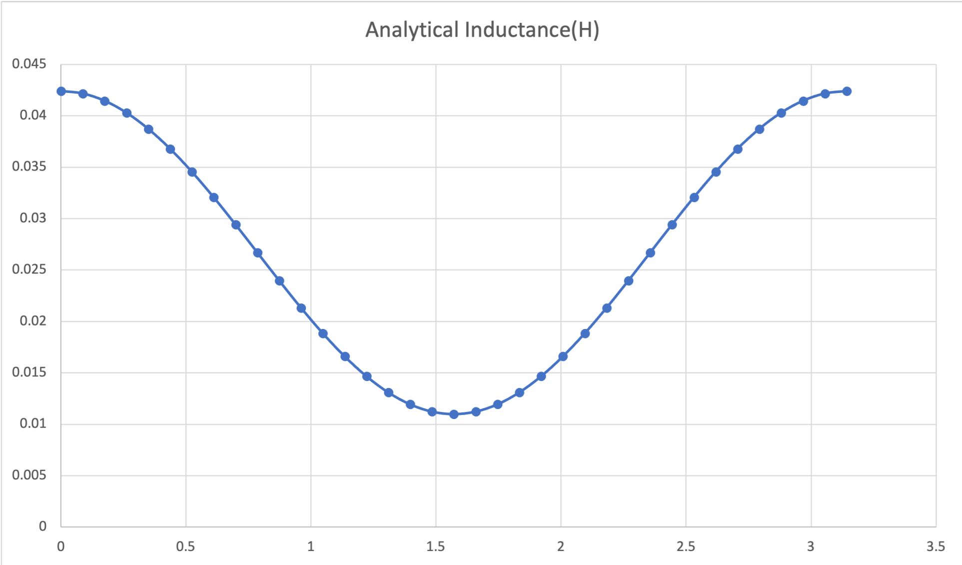

Figure 3: Analytical inductance variation as a function of rotor angle

3.2.2 Torque Versus Angle Plot

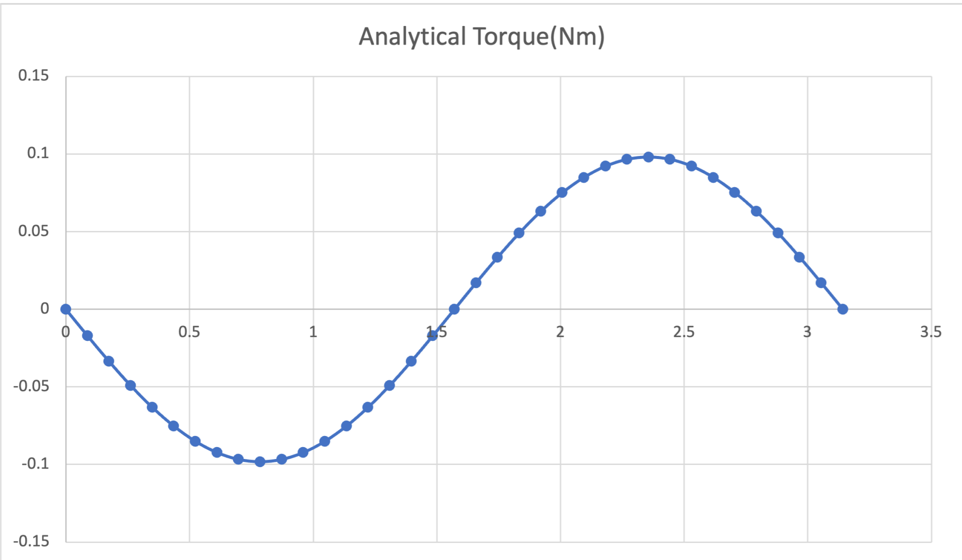

Figure 4: Analytical torque as a function of rotor angle under constant DC excitation

3.3 Part c - Model Improvement Suggestions

- Fringing flux: In reality there is flux lines that go outward and they eventually can increase the air-gap cross-sectional area. So they can be modeled too.
- Rectangular cross-section: The assumption of area is rectangle can be changed with actual airg gap area which has curve.

- Finite core permeability: Assumption of finite permeability can be changed with actual electrical steel permeability.
- Saturation effects: Actual B-H curve can be used to increase actuality of calculations.
- Position-dependent effective area: The assumption of it is sinusoidal is not exactly true and can be calculated piecewise.

4 Q2 - 2D FEA Modelling with Linear Materials

4.1 Software and Modelling Approach

For the FEA Modelling I used the COMSOL Multiphysics.

4.2 Material Selection

For this part i choosed M330-35A electrical steel with the properties below.

Table 2: Properties of M330-35A Electrical Steel

Property	**Value**
Grade	M330-35A (CRNGO)
Specific core loss (1.5 T, 50 Hz)	$\leq$ 3.30 W/kg
Relative permeability (μ_r)	7000 (assumed)

4.3 Design

We can see the 2D design of this project in below figure. It has air definition as an outer box with magnetic insulation in borders. Also design has rotating air gap around its rotor.

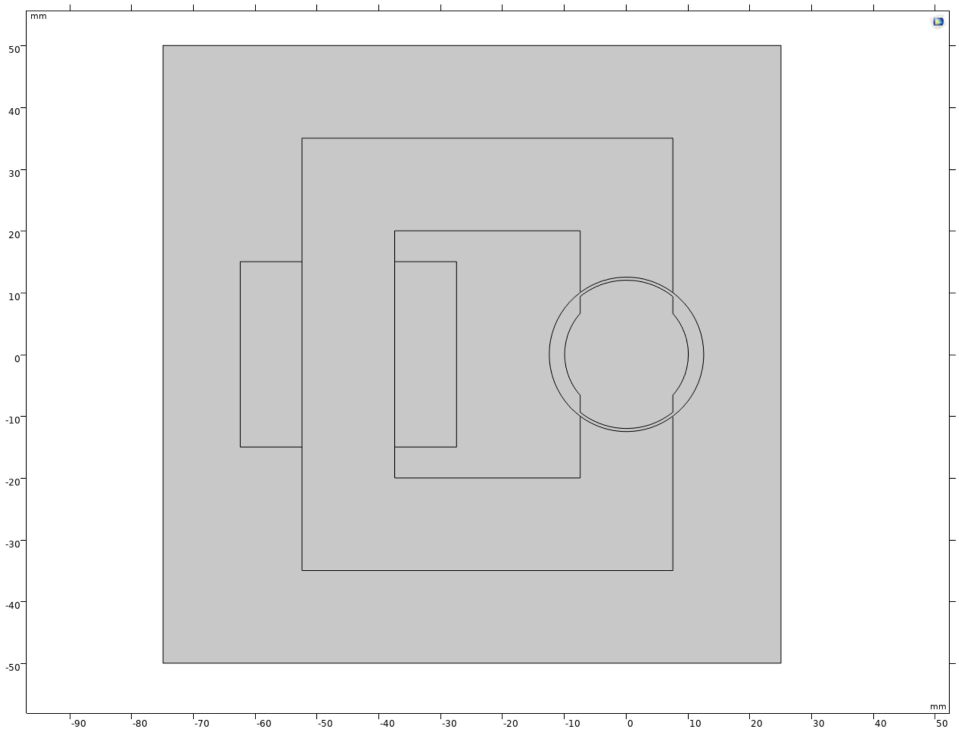

Figure 5: 2D design of the motor

4.4 Mesh Settings

I choosed the exteremly fine mesh settings for better results. By giving some tradeoff in computation time. We can see the meshes of the design in below figure.

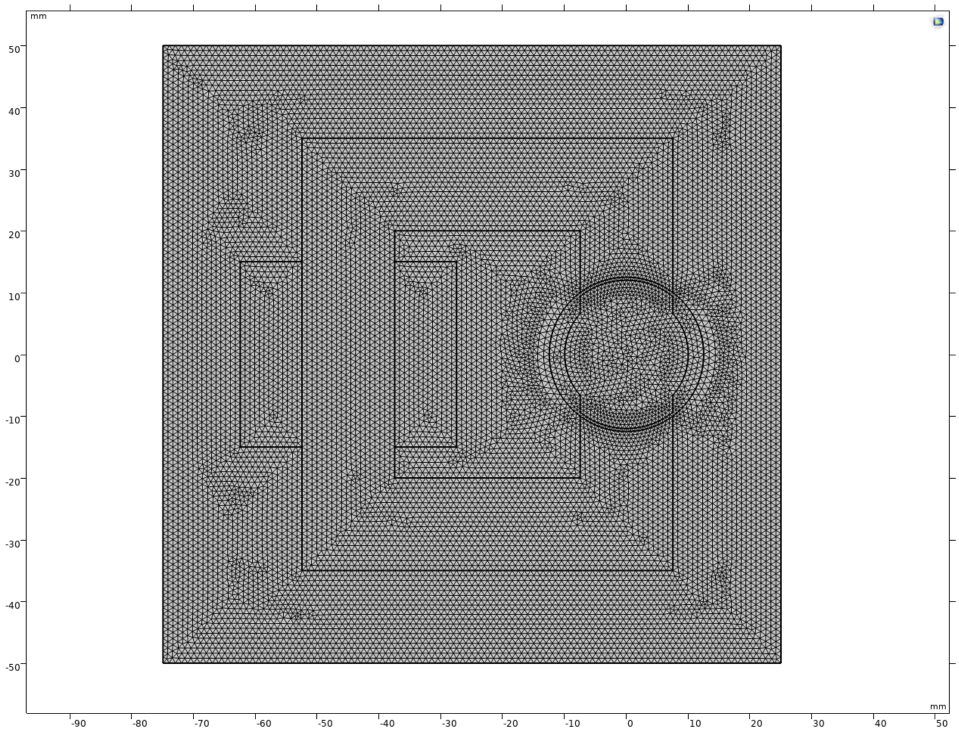

Figure 6: Meshes of the 2D design of the motor

4.5 Part a - Flux Density Distributions at Selected Positions

Readable flux density vector plots at 0°, 45°, and 90° are shown in Figures 7, 8, and 9.

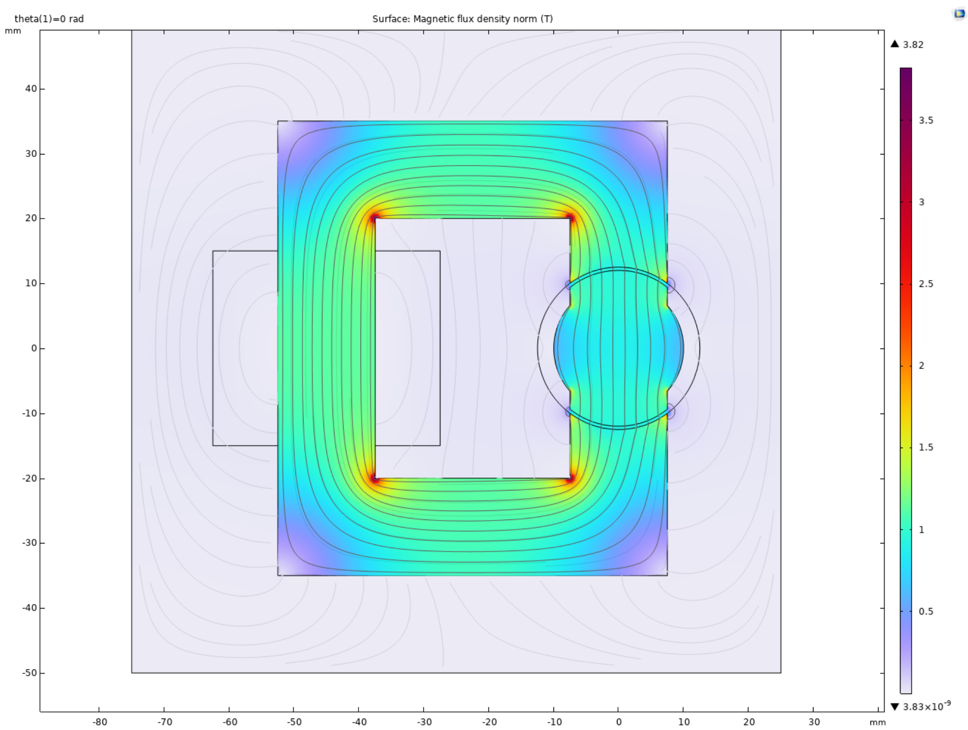

Figure 7: Flux density vector plot for the linear material model, $\theta = 0°$.

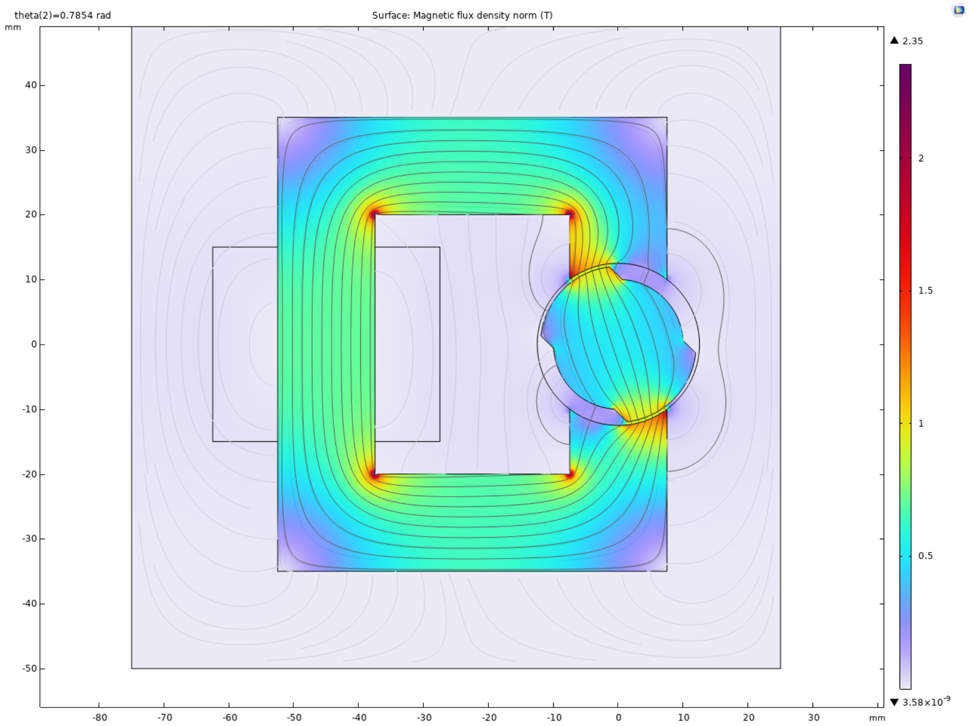

Figure 8: Flux density vector plot for the linear material model, $\theta = 45°$.

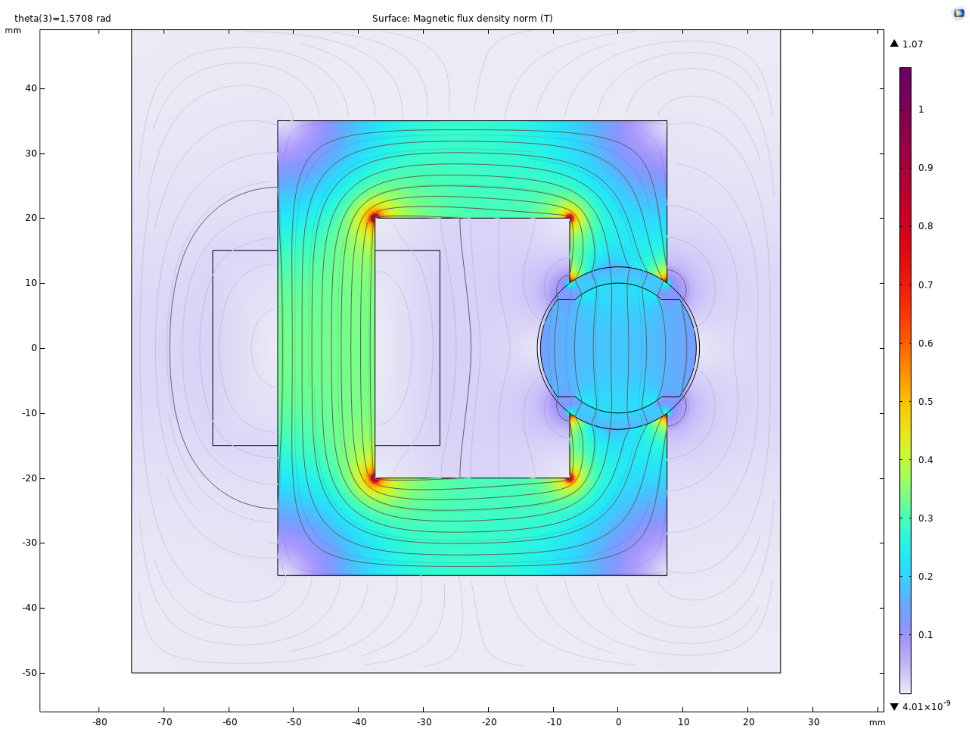

Figure 9: Flux density vector plot for the linear material model, $\theta = 90°$.

4.6 Part b - Inductance and Stored Energy Results

Table 3: Linear FEA results at selected rotor positions

Position (deg)	Inductance (H)	Stored Energy (J)
0	0.05	0.157
45	0.031	0.097
90	0.015	0.046

4.7 Part c - Torque Versus Angle

For torque analysis I made parametric sweep from 0 degree to 180 degree with 5 degree increment. And get the torque result from COMSOL using force calculation. We can see the resulting torque versus angle plot below.

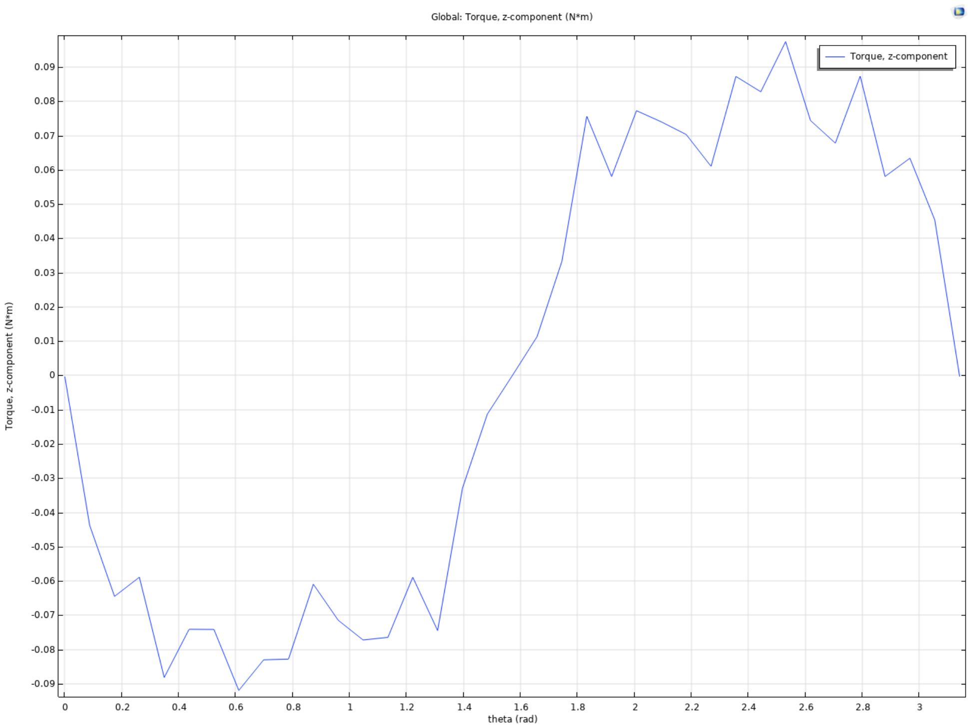

Figure 10: Torque versus angle obtained from 2D linear FEA

4.8 Comparison with Analytical Model

When we get the total magnetic energy value from COMSOL for each theta value I calculate inductance using it. Below we can see the resulting Analytical and FEA based inductance comparison graph.

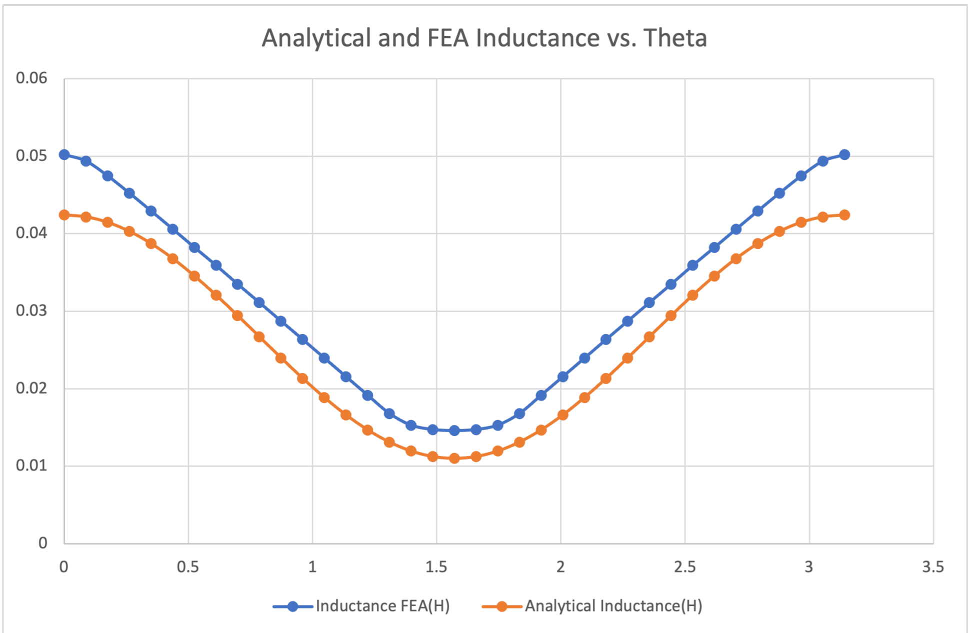

Figure 11: Analytical and FEA based inductance versus angle plot

And comparison plot of analytical and FEA based torque versus angle plot can be seen below:

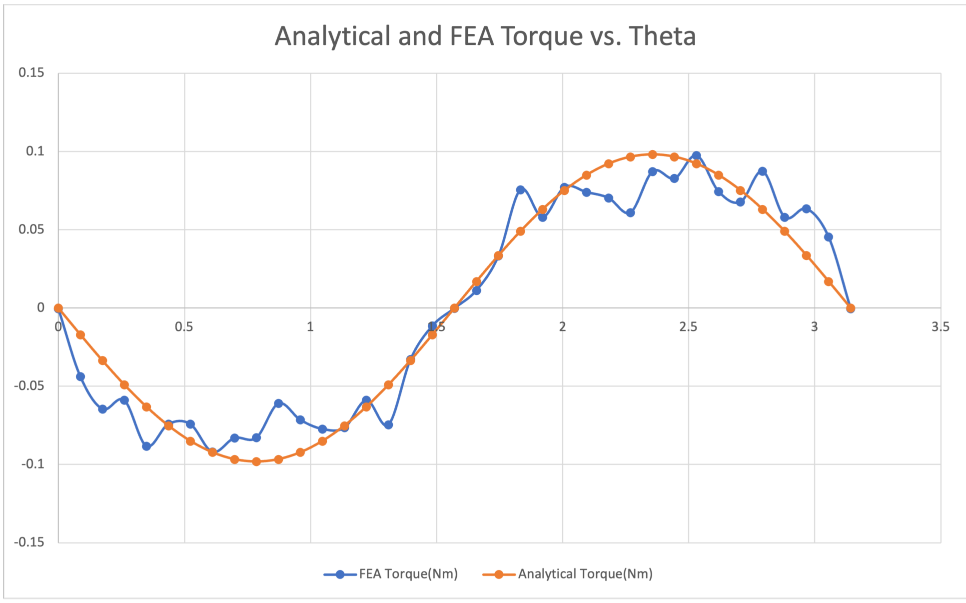

Figure 12: Analytical and FEA based torque versus angle plot

5 Q3 - 2D FEA Modelling with Nonlinear Materials

5.1 Nonlinear Material Model

For this part I used BH curve of the M330-35A from SIWATT and COMSOL builtin Soft Iron. I compare two nonlinear material because the material I choose has large value for saturated flux density. In the figure below we can see the BH curves.

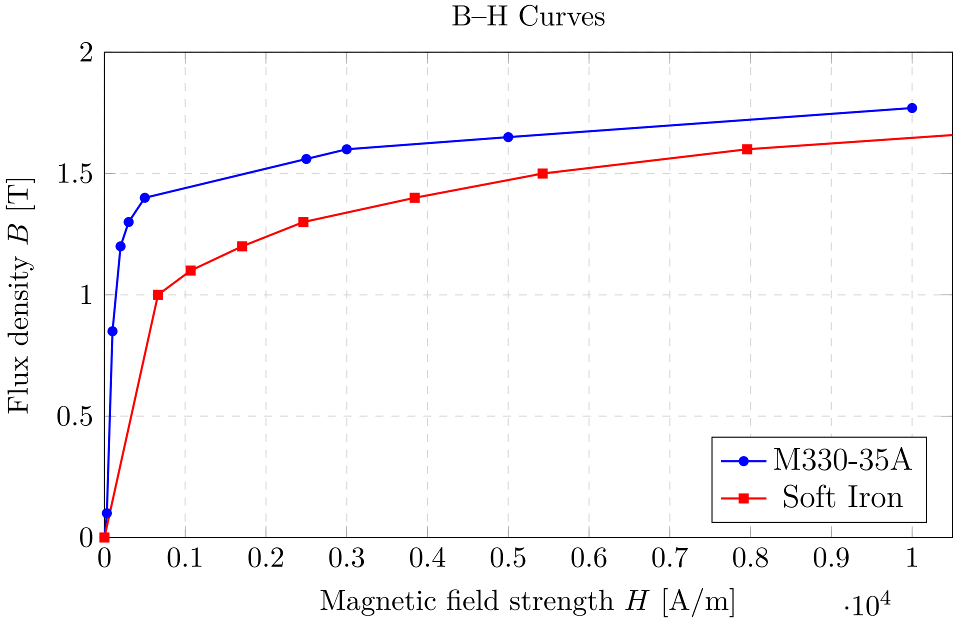

Figure 13: Nonlinear B–H characteristics of M330-35A electrical steel and Soft Iron.

5.2 Analysis of M330-35A

5.2.1 Part a - Flux Density Distributions at Selected Positions

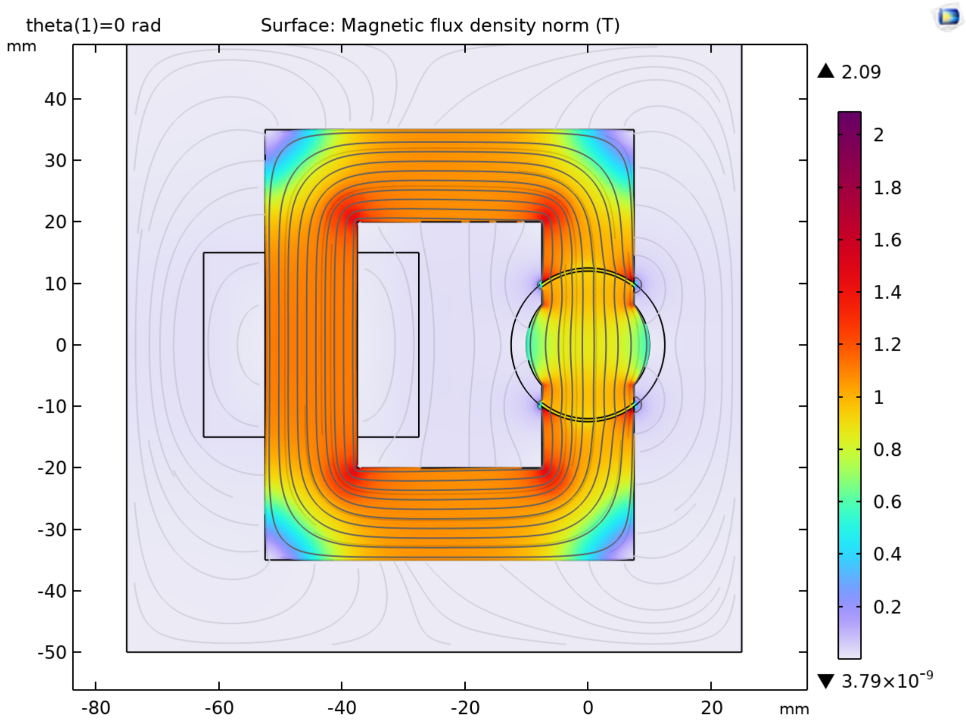

Figure 14: Flux density vector plot for the linear material model, $\theta = 0°$.

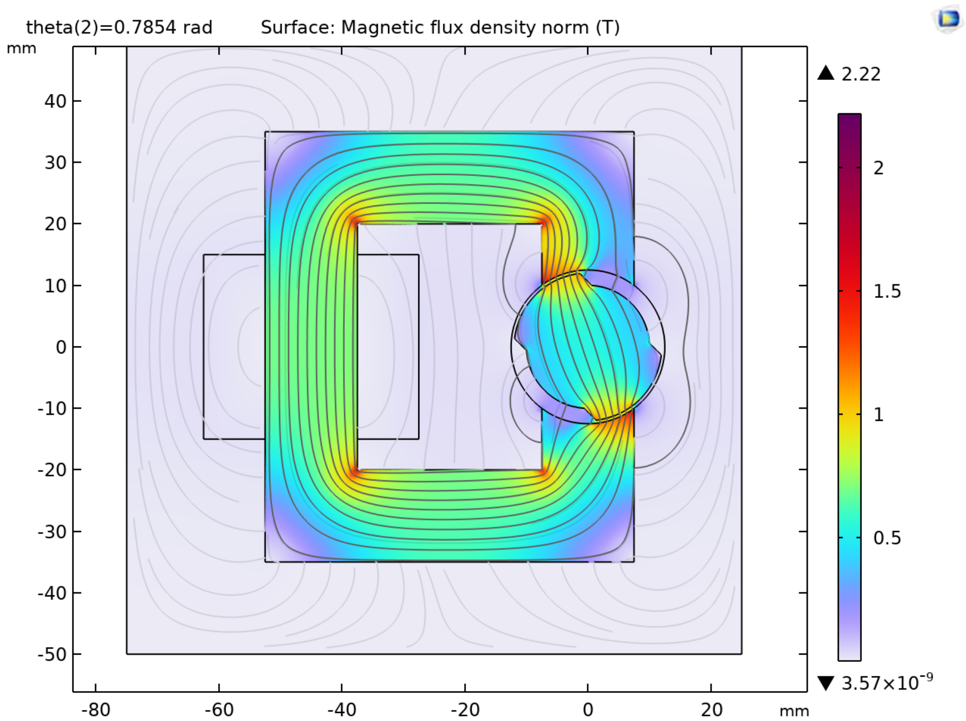

Figure 15: Flux density vector plot for the linear material model, $\theta = 45°$.

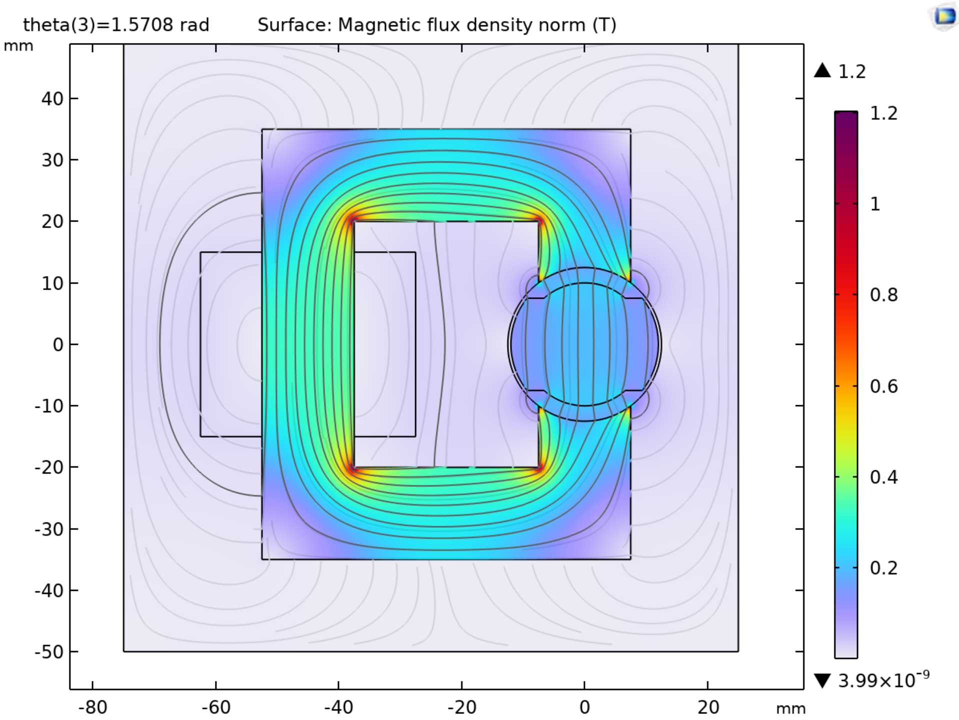

Figure 16: Flux density vector plot for the linear material model, $\theta = 90°$.

5.2.2 Part b - Inductance and Energy Results

Table 4: Nonlinear FEA results at selected rotor positions

Position (deg)	Inductance (H)	Stored Energy (J)
0	0.049	0.154
45	0.031	0.097
90	0.015	0.046

5.2.3 Part c - Torque versus Angle

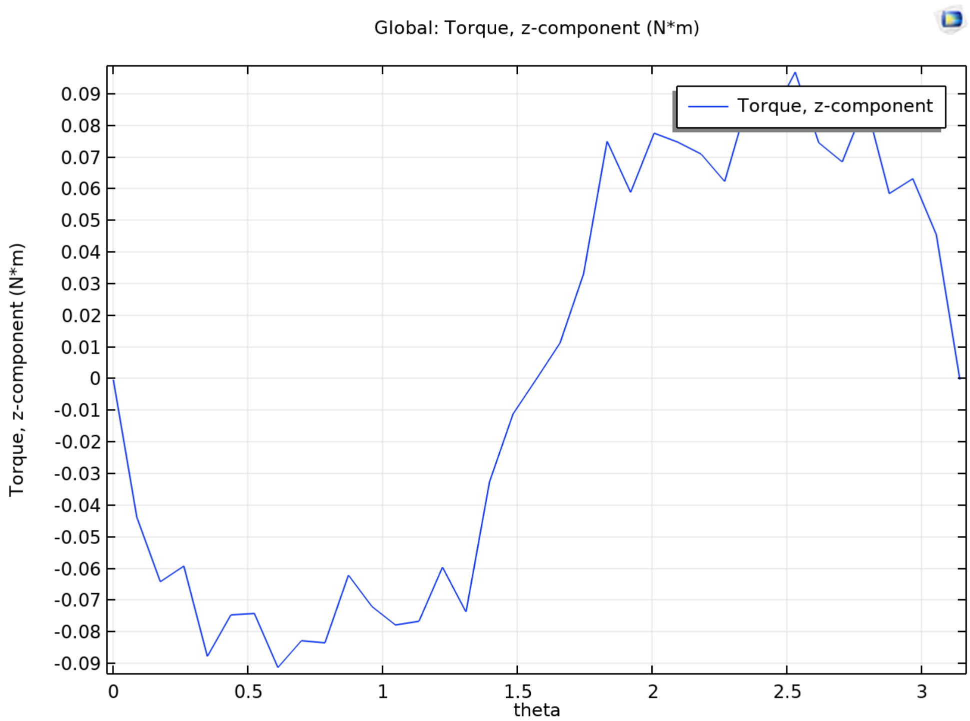

Figure 17: Torque versus angle obtained from 2D nonlinear FEA

5.3 Analysis of Soft Iron

5.3.1 Part a - Flux Density Distributions at Selected Positions

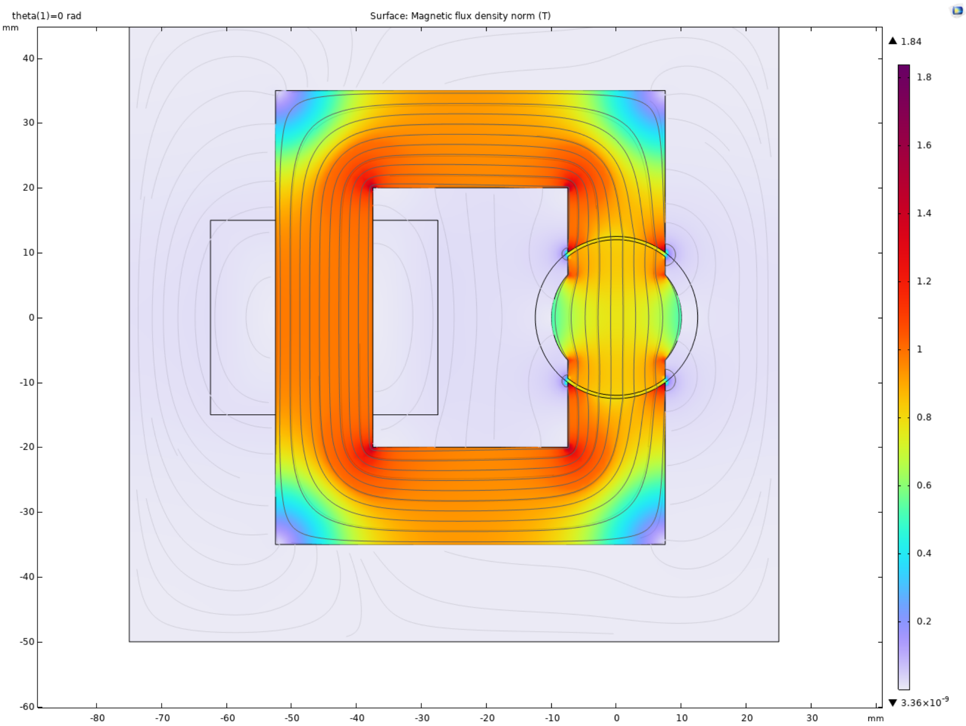

Figure 18: Flux density vector plot for the linear material model, $\theta = 0°$.

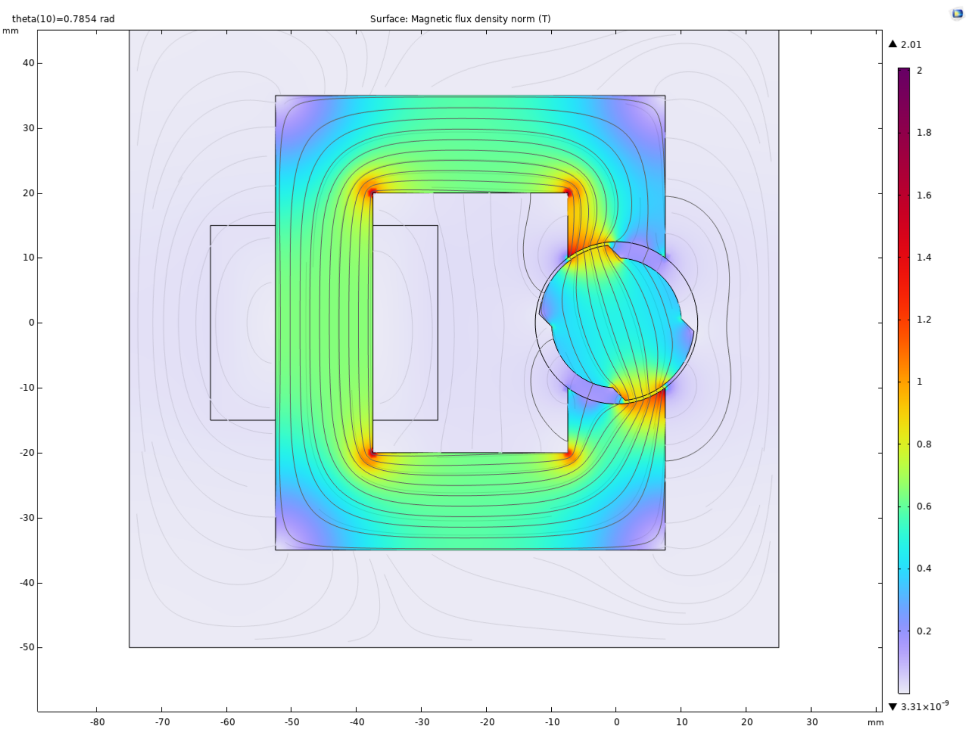

Figure 19: Flux density vector plot for the linear material model, $\theta = 45°$.

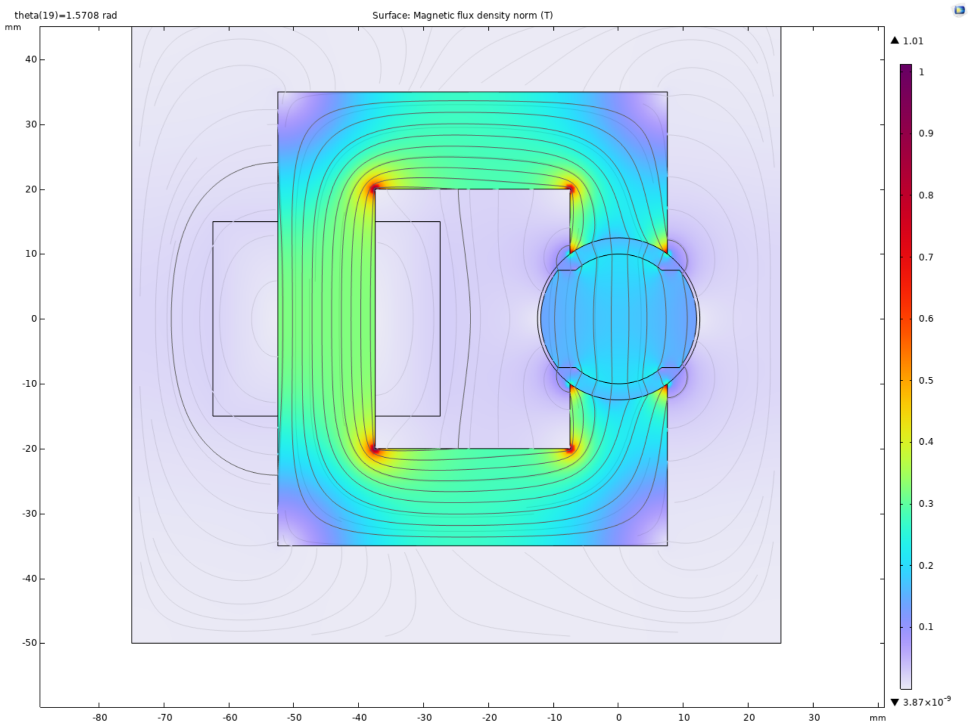

Figure 20: Flux density vector plot for the linear material model, $\theta = 90°$.

5.3.2 Part b - Inductance and Energy Results

Table 5: Nonlinear FEA results at selected rotor positions

Position (deg)	Inductance (H)	Stored Energy (J)
0	0.044	0.136
45	0.29	0.089
90	0.014	0.044

5.3.3 Part c - Torque versus Angle

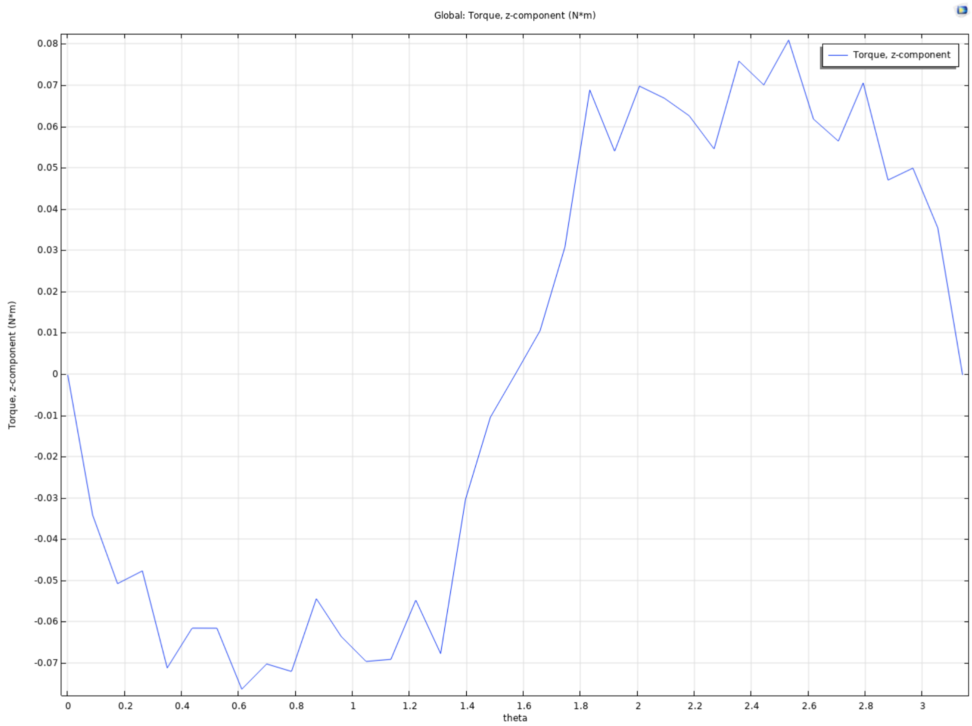

Figure 21: Torque versus angle obtained from 2D nonlinear FEA

5.4 Comparison with Analytical Model

When I do the same comparison that I did with the linear model I get the below figures. From these figures we can see that the inductance is higher than analytical assumptions. And when we compare between M330-35A and soft iron we can say that due to early saturation and smaller value saturation of soft iron we see less inductance in full alligned position, e.g 0 degree, because this is the position where we see higher flux density thus saturation.

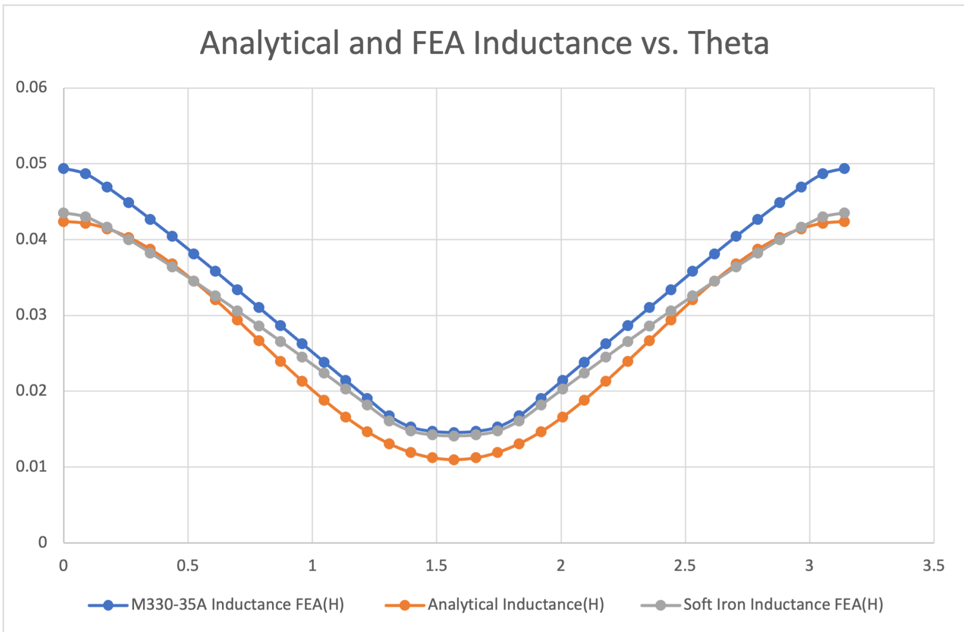

Figure 22: Analytical and FEA based inductance versus angle plot

And comparison plot of analytical and FEA based torque versus angle plot can be seen below. In this one we can see that soft iron produce less torque than M330-35A and analytical assumption create more torque when unalligned and less torque when alligned than both FEA analysis.

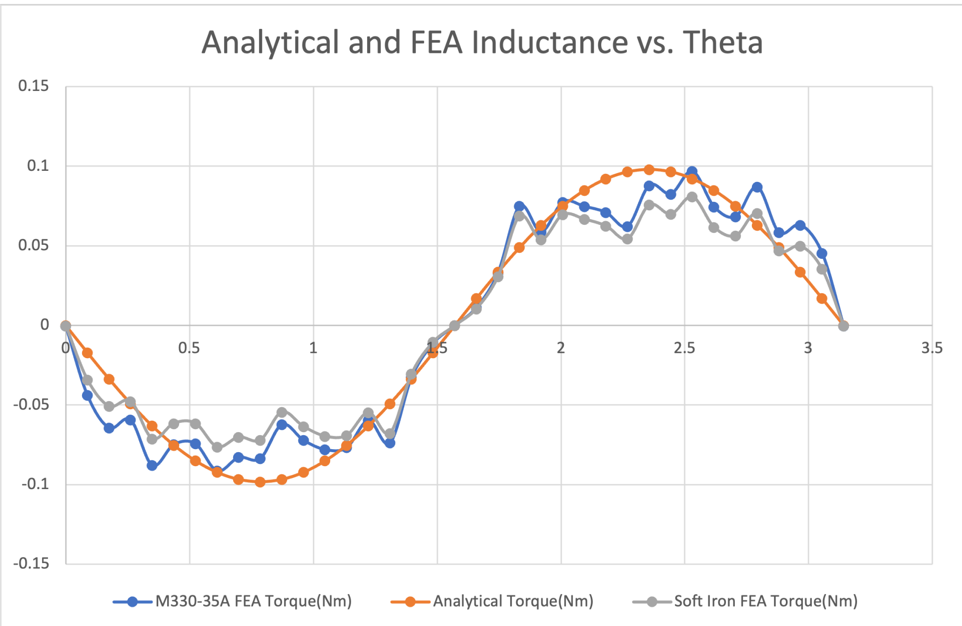

Figure 23: Analytical and FEA based torque versus angle plot

5.5 Part d - Comparison of Linear, Nonlinear and Analytical Results

In below table we can say that linear model is highly optimistic model and M330-35A barely saturates so we do not see much difference between it and linear one. But in soft iron we see up to 15% differences.

Table 6: Comparison of analytical, linear FEA, and nonlinear FEA inductance results

Position (deg)	L_{ana}	L_{lin}	$L_{\text{nonlin}M330-35A}$	$L_{\text{nonlin}SoftIron}$
0	0.0424	0.05	0.049	0.044
45	0.0267	0.031	0.031	0.29
90	0.011	0.015	0.015	0.014

In below table we can see that analytical calculation gave higher torque but in linear and M330-35A we see same torque up to 2 figures. And in soft iron we see less torque.

Table 7: Comparison of analytical, linear FEA, and nonlinear FEA torque results

Position (deg)	T_{ana}	T_{lin}	$T_{\text{nonlin}M330-35A}$	$T_{\text{nonlin}SoftIron}$
0	0	0	0	0
45	-0.1	-0.08	-0.08	-0.07
90	0	0	0	0
135	0.1	0.09	0.09	0.08
180	0	0	0	0

5.6 Part d - Discussion of Fringing and Saturation Effects

- When core reach to the knee of B-H curve incremental permeability of core drop sharply that is why soft iron has 0.044H inductance while linear core has 0.05H, M330-35A barely saturates at 2.5 A, so its results stay close to the linear case.
- Torque depends on derivative of inductance with respect to theta in Soft Iron we can see it is more flattened near aligned position thus it decrease the torque.
- Fringing fluxes increase the effective air-gap area thus it increase the inductance specially in unaligned position. This is why we see it higher than analytical values.
- Since nonlinear system saturates, specially in soft iron, it changes the behavior of system as a result stored energy, inductance and torque values.

- The analytical model overestimates torque at low-overlap angles (because it assumes no leakage) and underestimates inductance at the unaligned position (because it ignores fringing). FEA captures both effects naturally.

6 Q4 – Control Method for Continuous Rotation

6.1 Working Principle

From the analytical model in Q1, the inductance of the coil as a function of rotor angle is:

$$L(\theta) \;=\; L_{avg} + L_{amp}\cos(2\theta), \tag{12}$$

with $L_{avg} = 0.0267\,\mathrm{H}$ and $L_{amp} = 0.0157\,\mathrm{H}$. The corresponding maximum and minimum inductances are $L_{\max} = L_{avg} + L_{amp} = 42.4\,\mathrm{mH}$ (aligned position) and $L_{\min} = L_{avg} - L_{amp} = 11.0\,\mathrm{mH}$ (unaligned position).

The electromagnetic torque developed by a singly-excited magnetically linear VR machine is obtained from the co-energy derivative:

$$T_e(\theta) \;=\; \frac{dW_m(\theta)}{d\theta} \;=\; \frac{d}{d\theta}\Big[\tfrac{1}{2}\Big(L_{avg} + L_{amp}\cos(2\theta)\Big)i^2\Big] \;=\; -\,i^2\,L_{amp}\,\sin(2\theta). \tag{13}$$

Because the torque is proportional to i^2, its *sign* is determined solely by the sign of $-\sin(2\theta)$. Over one mechanical revolution $[0, 2\pi)$, this quantity is positive on $\theta \in (90°, 180°) \cup (270°, 360°)$ (rising inductance, $dL/d\theta > 0$) and negative on $\theta \in (0°, 90°) \cup (180°, 270°)$ (falling inductance, $dL/d\theta < 0$). With a constant DC excitation the two halves cancel each other and the net average torque over one revolution is zero — the rotor cannot sustain rotation.

6.2 Proposed Control Strategy

To obtain a non-zero average torque, the coil current is gated synchronously with rotor position so that current flows *only while the inductance is rising*:

$$i(\theta) \;=\; \begin{cases} I_{\mathrm{DC}} = 2.5\,\mathrm{A}, & \text{if } \dfrac{dL}{d\theta} > 0 \;\; (-\sin 2\theta > 0), \\ 0, & \text{if } \dfrac{dL}{d\theta} \le 0. \end{cases} \tag{14}$$

This is a **unipolar pulsed-DC (position-synchronised ON/OFF)** excitation. Physically, while $dL/d\theta > 0$ the rotor pole is approaching alignment with the excited stator pole; the reluctance of the magnetic circuit is decreasing and the stored magnetic energy pulls the rotor towards the aligned position, producing a positive (motoring) torque. De-energising the coil before $dL/d\theta$ changes sign prevents the generation of negative (braking) torque as the rotor leaves the aligned position.

Substituting Eq. (14) into Eq. (13) gives the controlled torque

$$T_{\text{ctrl}}(\theta) \;=\; \begin{cases} -\, I_{\text{DC}}^2\, L_{amp}\, \sin(2\theta), & dL/d\theta > 0, \\ 0, & \text{otherwise}, \end{cases} \tag{15}$$

which is non-negative for every $\theta \in [0, 2\pi)$.

Average torque

Because the ON-region consists of two intervals of width $\pi/2$ each, the average torque over one mechanical revolution is

$$T_{\text{avg}} = \frac{1}{2\pi} \int_{\text{ON}} \Big(-I_{\text{DC}}^2\, L_{amp} \sin 2\theta\Big)\, d\theta = \frac{I_{\text{DC}}^2\, L_{amp}}{\pi}. \tag{16}$$

Substituting the numerical values,

$$T_{\text{avg}} = \frac{(2.5)^2 \cdot 0.0157}{\pi} \approx 31.2\,\text{mN m}. \tag{17}$$

The peak instantaneous torque (at $\theta = 135^\circ$ and $\theta = 315^\circ$) reaches $T_{\text{peak}} = I_{\text{DC}}^2 L_{amp} = 98.1\,\text{mN m}$.

6.3 Analytical and Numerical Results

Figure 24 shows the inductance profile, the gated coil current, and the resulting torque waveform for one full mechanical revolution. The green shaded zones represent the zones where the inductance is increasing and the coil is being excited. The grey dashed line represents the torque that results from leaving the coil on all the time; it is symmetrical about the origin, and thus it provides no net rotational effect. The blue line represents the torque achieved using the control scheme proposed, i.e., two positive spikes per revolution. The dotted horizontal line marks the average torque computed from Eq. (16).

Figure 25 compares the average torque produced over one revolution for the two cases. The always-ON DC excitation yields an average of essentially zero, whereas the proposed pulsed scheme produces 31.2 mN m of mean torque, confirming that continuous rotation can be sustained.

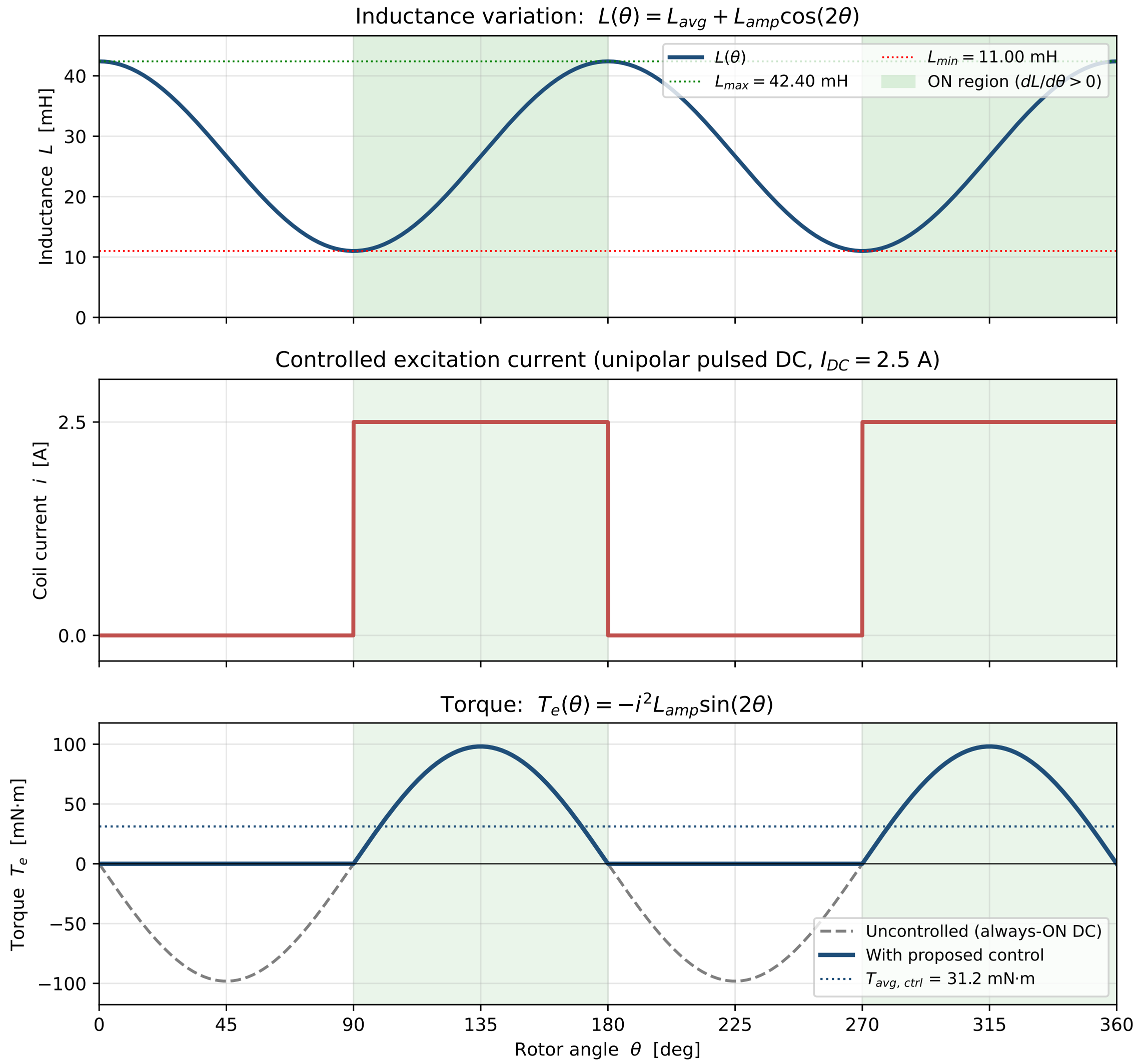

Figure 24: Proposed position-synchronised pulsed-DC control.

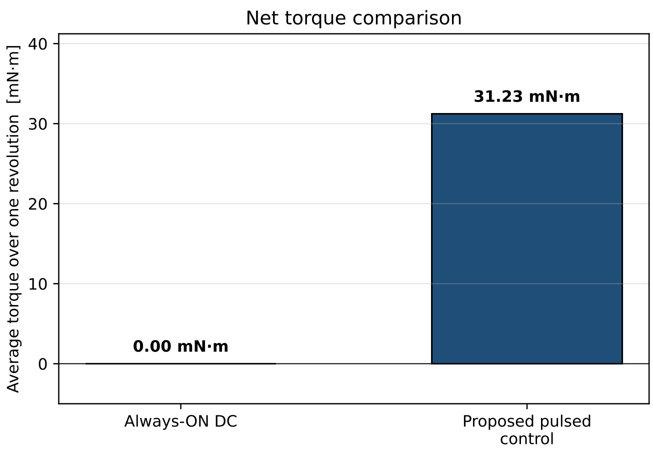

Figure 25: Average torque over one full mechanical revolution for the uncontrolled (always-ON) excitation versus the proposed position-synchronised pulsed-DC control.

6.4 Validation with the 2D FEA Model

The same control law was applied to the non-linear 2D FEA model of Q3 by giving 2.5A current only when theta between 90 degree to 180 degree in 0 to 180 degree simulation. The FEA results confirm the analytical prediction – the controlled torque envelope is non-negative for all θ.

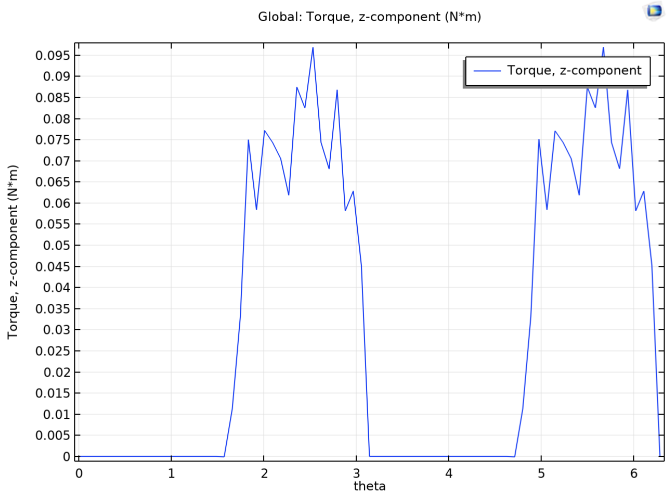

Figure 26: 2D FEA torque result for the proposed control method

7 Q5 - Animation of FEA Analysis

Animation of 2D FEA, with the proposed control method, can be found in GitHub Repository under Q5 folder.

8 Q6 - 3D FEA Analysis

For 3D FEA analysis again COMSOL Multiphysics used.

8.1 3D Geometry Figures

Designed 3D model can be seen below.

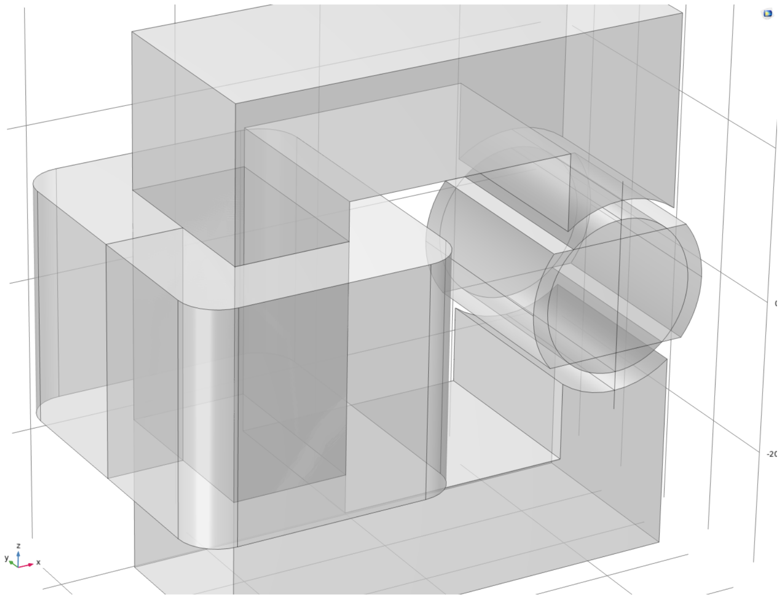

Figure 27: Side view of machine geometry

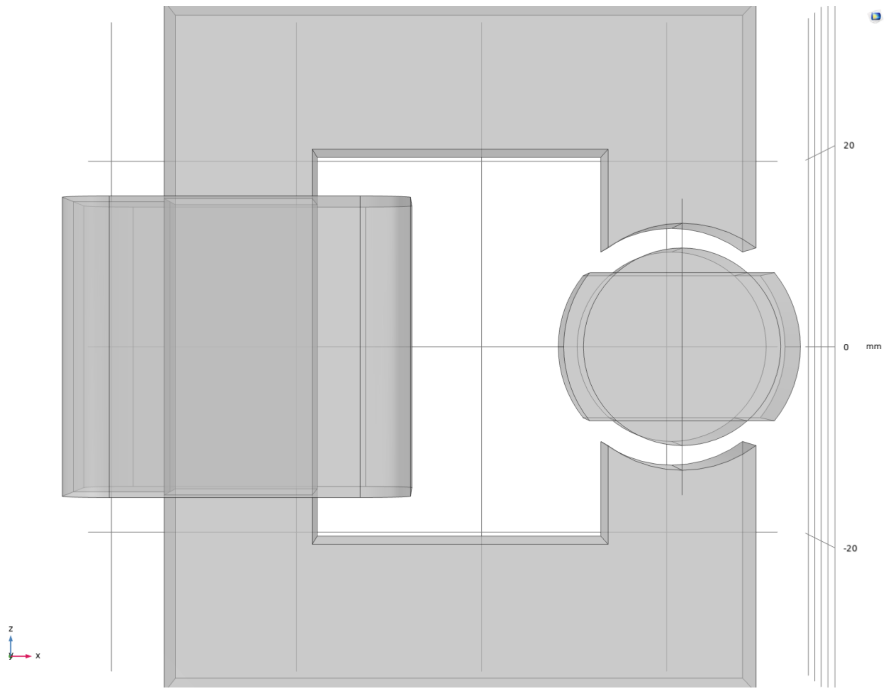

Figure 28: Direct view of machine geometry

8.2 Flux Density Distributions at Selected Positions

Flux density vector plots at 0°, 45°, and 90° are shown in Figures 29, 30, and 31.

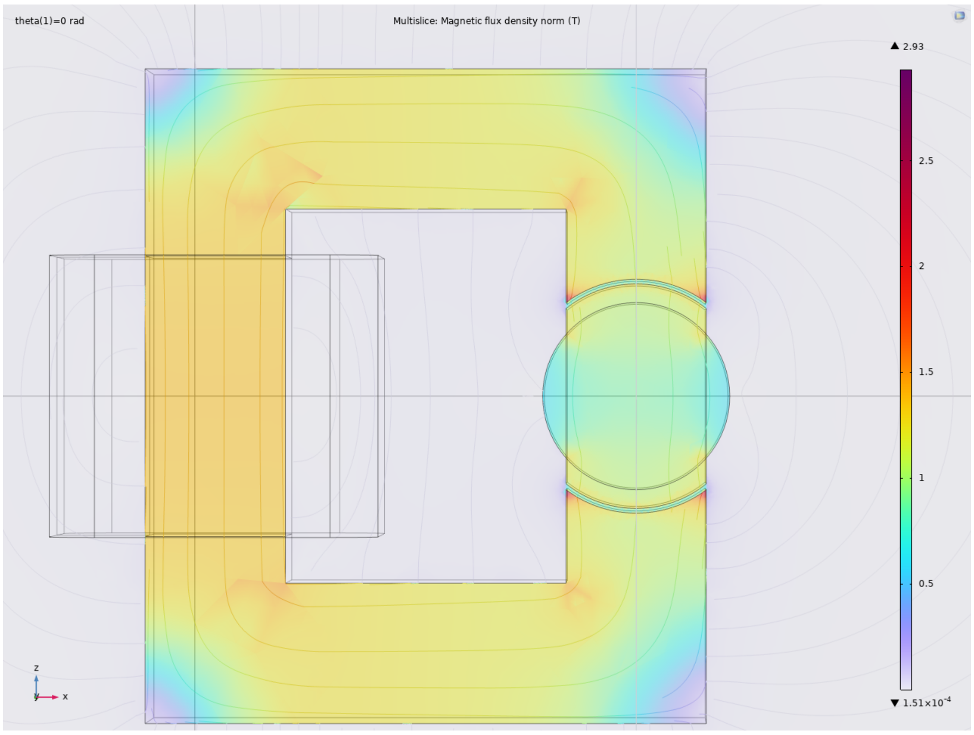

Figure 29: Flux density vector plot for the 3D model, $\theta = 0°$.

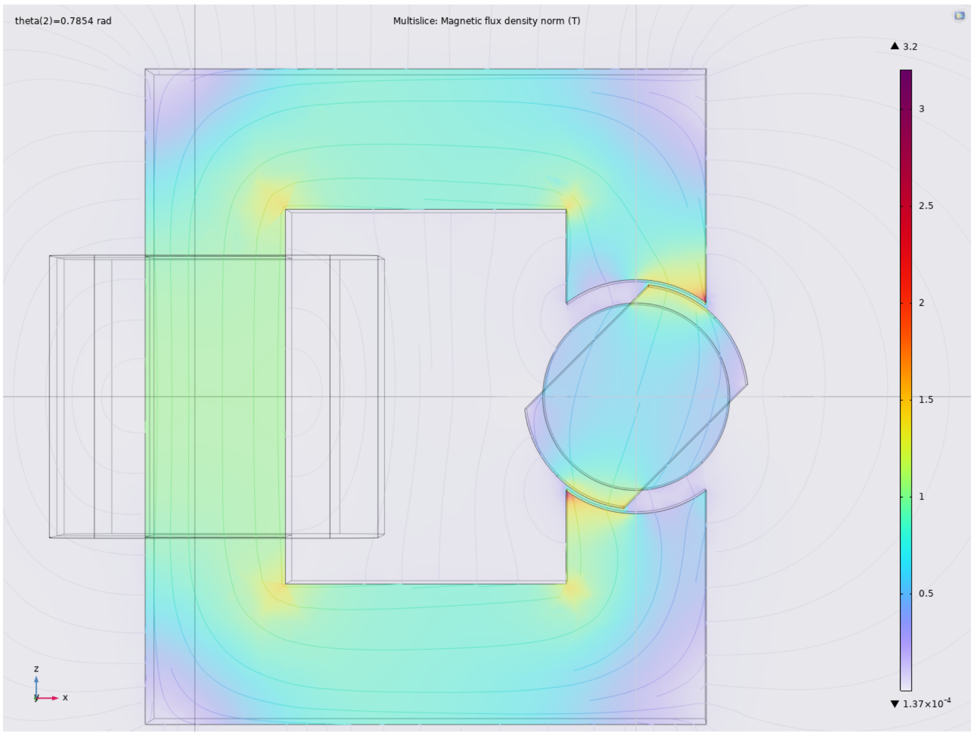

Figure 30: Flux density vector plot for the 3D model, $\theta = 45°$.

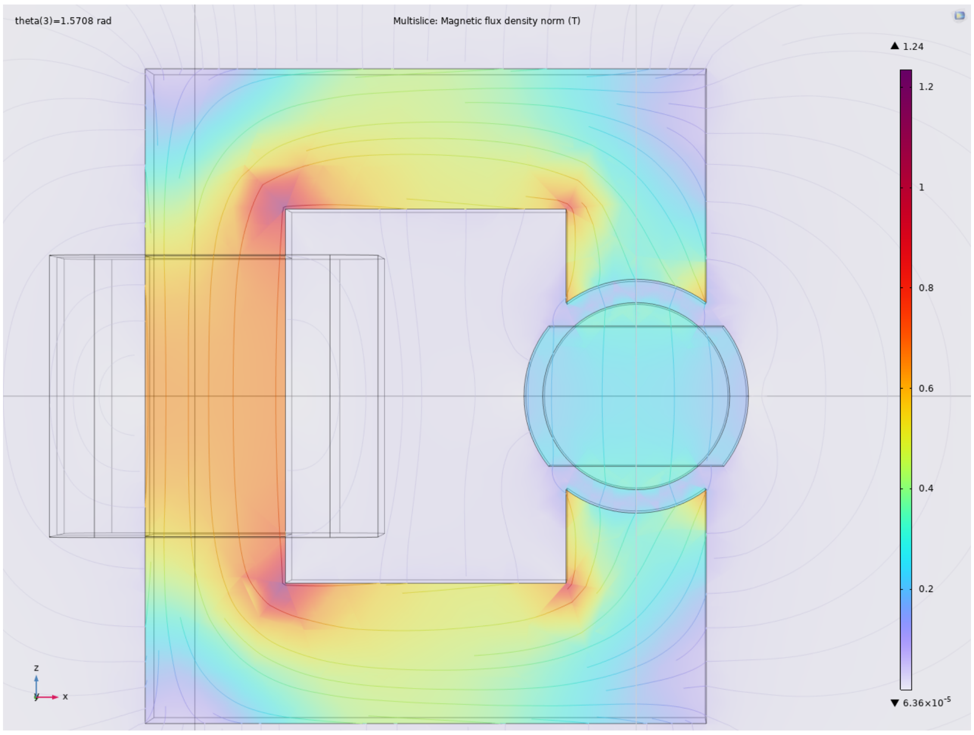

Figure 31: Flux density vector plot for the 3D model, $\theta = 90°$.

8.3 Flux Density Arrows at Selected Positions From Side View

Flux density arrow plots at 0°, 45°, and 90° are shown in Figures 32, 33, and 34.

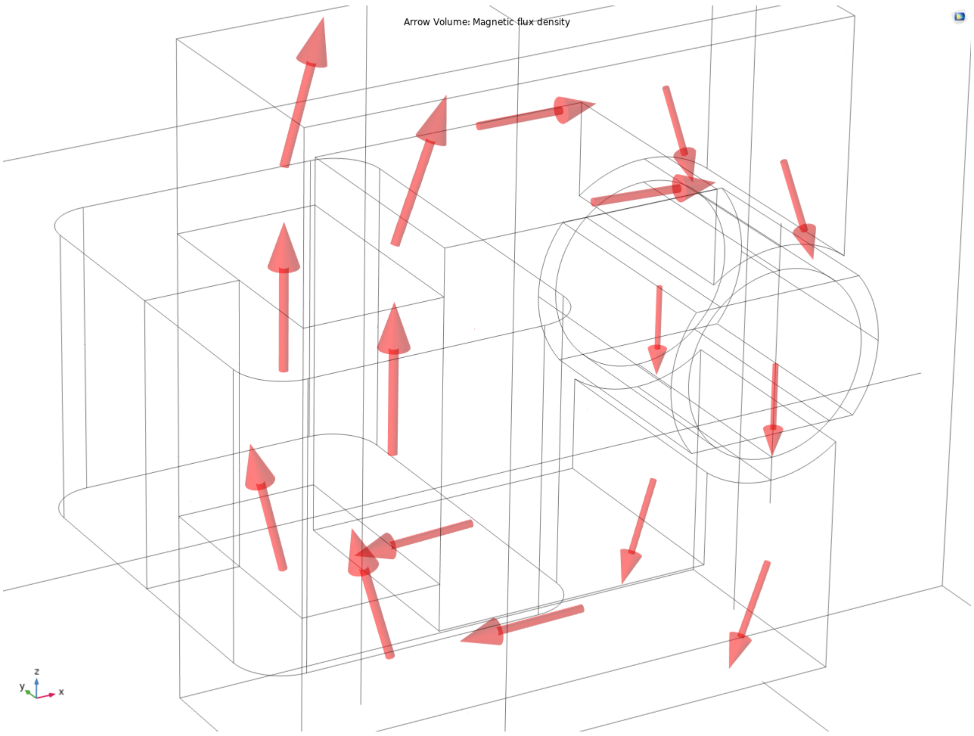

Figure 32: Flux density arrow plot for the 3D model, $\theta = 0°$.

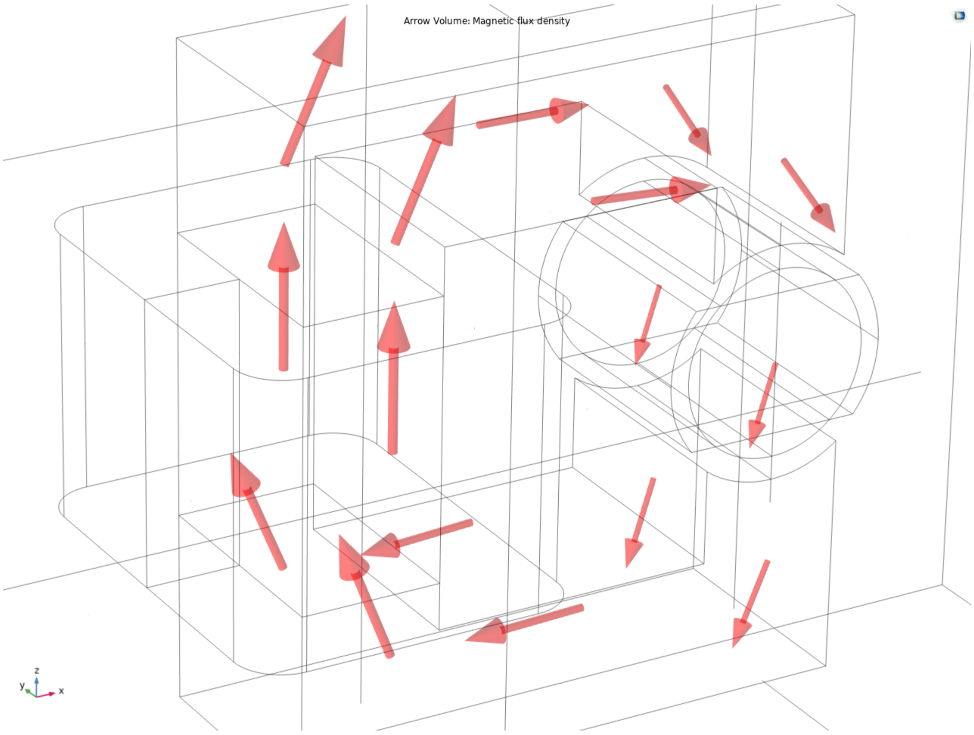

Figure 33: Flux density arrow plot for the 3D model, $\theta = 45°$.

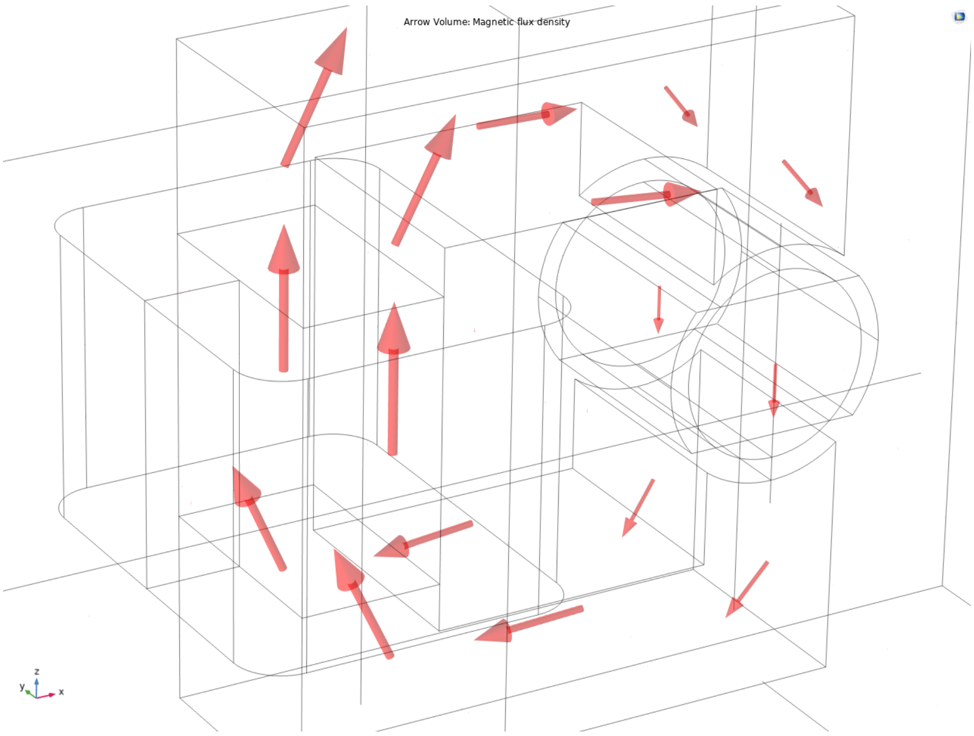

Figure 34: Flux density arrow plot for the 3D model, $\theta = 90°$.

8.4 Inductance and Energy Results and 2D FEA Comparison

Table 8: Nonlinear FEA inductance results at selected rotor positions

Position (deg)	Inductance 3D FEA (H)	Inductance 2D FEA (H)
0	0.06	0.049
45	0.045	0.031
90	0.03	0.015

Table 9: Nonlinear FEA stored energy results at selected rotor positions

Position (deg)	Stored Energy 3D FEA (J)	Stored Energy 2D FEA (J)
0	0.19	0.154
45	0.14	0.097
90	0.09	0.046

Probable cause of increase in stored energy and inductance is because of end windings which are neglected in 2D analysis. And also because of the axial fluxes which are also contributes to stored energy.

9 General Discussion

- Analytical, 2D linear FEA, and 2D nonlinear FEA models differ by no more than 10–15% in terms of inductance and maximum torque values. While the sinusoidal shape of inductance is generally predicted correctly, the FEA predicts broader peak values and higher L_min because of fringing effects.
- The assumption of infinite permeability, neglecting fringing and leakage fluxes lead to consistent underestimation of the minimum value of inductance and overestimation of maximum torque. The assumption of the rectangular air gap at an angle of 90° appears particularly unrealistic due to the rotor shape.
- Significance of nonlinearity modeling: While for M330-35A with the current of 2.5 A the difference is relatively small, Soft Iron shows up to 15% deviation. With greater currents or air gaps being smaller, modeling nonlinearity becomes crucial.
- 3D vs 2D: 3D analysis produced higher values of inductance because of end winding and axial fringes that cannot be predicted using 2D approach.

10 Conclusion

In this report, a variable reluctance machine was analyzed using three different approaches. The analytical solution was obtained, providing a maximum inductance of 42.4 mH, minimum inductance of 11.0 mH, and a sinusoidal torque with maximum values of ± 98 mN · m. Two-dimensional FEA results were obtained for linear material M330-35A, proving that while the shapes of inductance and torque are close to those in the analytical solution, inductance for unaligned position is higher because of fringing that cannot be captured by the analytical solution. Nonlinear FEA for M330-35A resulted in only slight changes compared to the linear case as the material is slightly saturated at current 2.5 A; however, the Soft Iron material provided up to 15% decrease in torque and inductance due to lower saturation flux density. A position-synchronous pulsed DC control was investigated and resulted in positive torque with average value 31.2 mN · m. Three-dimensional FEA provided higher inductance and magnetic energy storage than 2D due to the effect of end-winding and axial-fringing ignored in 2D. The main takeaways are that simple analytical models are useful for first-order sizing, that fringing and saturation are significant and always act to reduce the inductance slope, and that even the simplest single-phase VR machine requires position-synchronised excitation to produce net torque.